

OS Question Bank for Periodic Test II
SE (Computer Engineering) -NEP
Class: C1, C2, C3

Note: In every explanation, diagram becomes mandatory wherever applicable.

1. Define the following terms :
 - a. Mutual Exclusion
 - b. Race Condition
 - c. Critical section
 - d. Concurrency
 - e. Primitives of Mutual exclusion
 - f. Busy wait
 - g. Seek time
 - h. Rotational delay
2. What are the requirements of a valid solution to the critical section problem.
3. Explain the two process solutions with examples and advantages and disadvantages
 1. Using turn variable
 2. Using State flag
 3. Peterson's solution.
4. Explain the hardware solutions to achieve Mutual Exclusion with examples and advantages and disadvantages (TSL , Interrupt enable/disable)
5. How to achieve mutual exclusion using semaphore (Binary and Counting semaphore) explain with examples,
6. What is the Producer Consumer problem . Explain with examples.
7. What are the requirements to be met to solve the producer Consumer problem using semaphore.
8. Explain the solution to Producer consumer problem using the appropriate code and semaphores and examples.
9. What is a deadlock? What are necessary conditions for a deadlock?
10. What is a Resource Allocation Graph? What is a Wait For Graph and when is it used?
11. Explain how deadlock can be prevented using necessary conditions
12. Deadlock Avoidance with Multiple instances of a single resource type and Multiple instances of multiple resource types (Banker's Algorithm, allied numerical)
13. Deadlock detection (Banker's Algorithm, allied numerical) and recovery from deadlocks.
14. Explain Mass Storage Structure with example
15. Disk scheduling algorithms and numericals
16. Concepts of Virtual memory, Paging, Logical address, Physical address, Page table, conversion of Logical address to physical address, demand paging,
17. Explain the use of Translation Lookaside Buffer (TLB) in detail with proper diagram?

18. Explain the need for page replacement algorithms.
19. What are the different page replacement algorithms , explain with advantages and disadvantages .
20. Numericals on page replacement algorithms.
21. Explain thrashing in detail.
22. Explain Segmentation with suitable diagram and example. (along with Numericals)
23. Differences between Paging and Segmentation.
24. Internal and External fragmentation w.r.t. paging and segmentation
25. Explain File organization techniques (with diagrams)
26. Explain File allocation techniques (with diagrams)
27. Short notes on - Network OS , Distributed OS, Open Source , Proprietary OS, RTOS and Cloud OS
28. Explain Copy-On write with example.
29. Explain Interrupt-Driven IO and Programmed IO with proper flowchart.
30. Explain DMA with proper diagram.